Assignment 2: Data Cleaning and Transformation

Dataset Report & Transformation Audit Log

TASK: CORRECTING INCONSISTENT DATA

- **Identify Inconsistent Text Formats:** Detect inconsistent text capitalization present in the *"Product Name"* column (e.g., lowercase `headphones`, `smartphone` vs. Proper Case).
- **Identify Typos in Category:** Locate misspellings and truncated text in the *"Category"* column (e.g., `Electronicscscs`, `Electroni`).
- **Standardize via Find & Replace:** Use Excel's Find and Replace functionality to standardize all product names to Proper Case and correct all category typos.

1. FIND & REPLACE OPERATIONS LOG

STEP	TARGET COLUMN	FIND WHAT	REPLACE WITH	PURPOSE / RESULT
1	Product Name	`headphones`	`Headphones`	Standardized lowercase product entries to Proper Title Case
2	Product Name	`smartphone`	`Smartphone`	Corrected lowercase smartphone entries to capitalized format
3	Category	`Electronicscscs` / `Electroni`	`Electronics`	Cleaned 6 corrupted/truncated category spelling errors across cells

2. COMPLETE DATASET (POST-TRANSFORMATION)

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY	PRICE RANGE	CATEGORY (CLEANED)
1	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	130	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics	900	Electronics
5	22-MAY-US	Backpack	North Face	70	20		70	Electronics
6	07-JUN-UK	Headphones	Sony		45	Electronics	309	Electronics
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40		90	Electronics
13	08-DEC-DE	Coffee Maker	Nespresso	120	35		120	Electronics
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	150	Electronics
15	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	160	Electronics
18	27-JAN-IT	Laptop	Asus	980	10	Electronics	980	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	150	Fashion
20	14-AUG-US	Camping Tent	Coleman		10	Outdoor	309	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics	700	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	80	Kitchen

#	PRODUCT ID	PRODUCT NAME	BRAND NAME	PRICE ($)	QUANTITY	CATEGORY	PRICE RANGE	CATEGORY (CLEANED)
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30		150	Electronics
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories
25	29-SEP-CA	Smartphone	Google	800	45	Electronics	800	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	309	Fashion
27	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	400	Kitchen
28	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics
29	07-MAR-CA	Dress	Zara	60	30	Fashion	60	Fashion
30	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	40	Kitchen
31	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	130	Electronics
32	02-DEC-CA	Jeans	Levi's	50	50	Fashion	50	Fashion
33	17-JUN-IN	Laptop	HP	950	25	Electronics	950	Electronics
34	09-JUL-FR	Watch	Casio	100	20	Accessories	100	Accessories